CS & IT ENGINEERING

THEORY OF COMPUTATION

Grammar

Lecture - 02

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Recap of Previous Lecture

Topic

1 Grammar

?????

2 $G=(N, T, P, S)$

Topics to be Covered

Topic $G = (N, T, P, S)$

Topic ?? Grammar Construction

Topic ?? derivation

Topic ?? $\left\{\begin{array}{l} L.M.D \\ R.M.D \end{array}\right\}$

Topic : Grammar

- Set of rules used to describe strings of a language is known as grammar.
- Formal definition of grammar is

Ex :

$S \to ABC$

$A \to a \mid b$

$B \to b$

$C \to a \mid d$

G=(N, T, P, S)

{S, A, B, C}

- **N** :- non terminals (or) variables = 4
- **T** :- Terminals → {a, b, d} ⇒ 3
- **P** :- no. of productions → 6
- **S** :- Starting symbol

Topic : Grammar

- For every language grammar exit & every grammar generates one language.
- All grammars is of a from α replacement of $\alpha \rightarrow \beta$, where β is replacement of α

Topic : Derivation

- The process of deriving strings from the given grammar known as derivation.
- The derivation can be either left most derivation or right most derivation

- **Left most derivation:**
  It is the derivation in which left most non terminal is replaced by its R.H.S part at every step.

- **Right most derivation:**
  It is a derivation in which right most non terminal is replaced by its R.H.S part at every step.

Ex:-

L.M.D

$$S$$
$$\boxed{ABC}$$
$$\Downarrow$$
$$aBC$$
$$\downarrow$$
$$abC$$
$$\Downarrow$$
$$abd$$

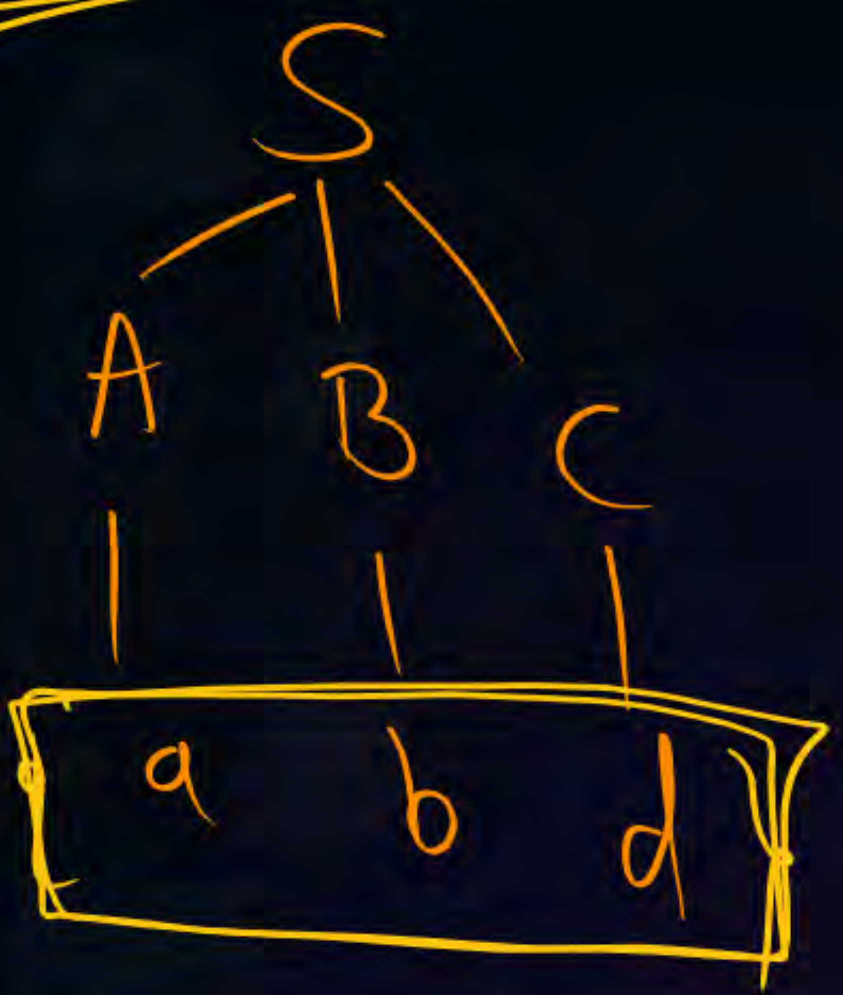
R.M.D

$$S$$
$$ABC$$
$$\Downarrow$$
$$ABd$$
$$\Downarrow$$
$$Abd$$
$$abd$$

Ex:-

$$\boxed{\begin{array}{l} S \to ABC \\ A \to a \\ B \to b \\ C \to d \end{array}}$$

Derivation tree



Derivation Tree (or)Parse Tree

- Tree representation of the derivation is known as derivation tree.
- All leaf node of the parse tree is known as yield of parse tree .
- while reading yield from left to right sentence of the grammar can be generate.

→ For Regular Language → Regular Grammar

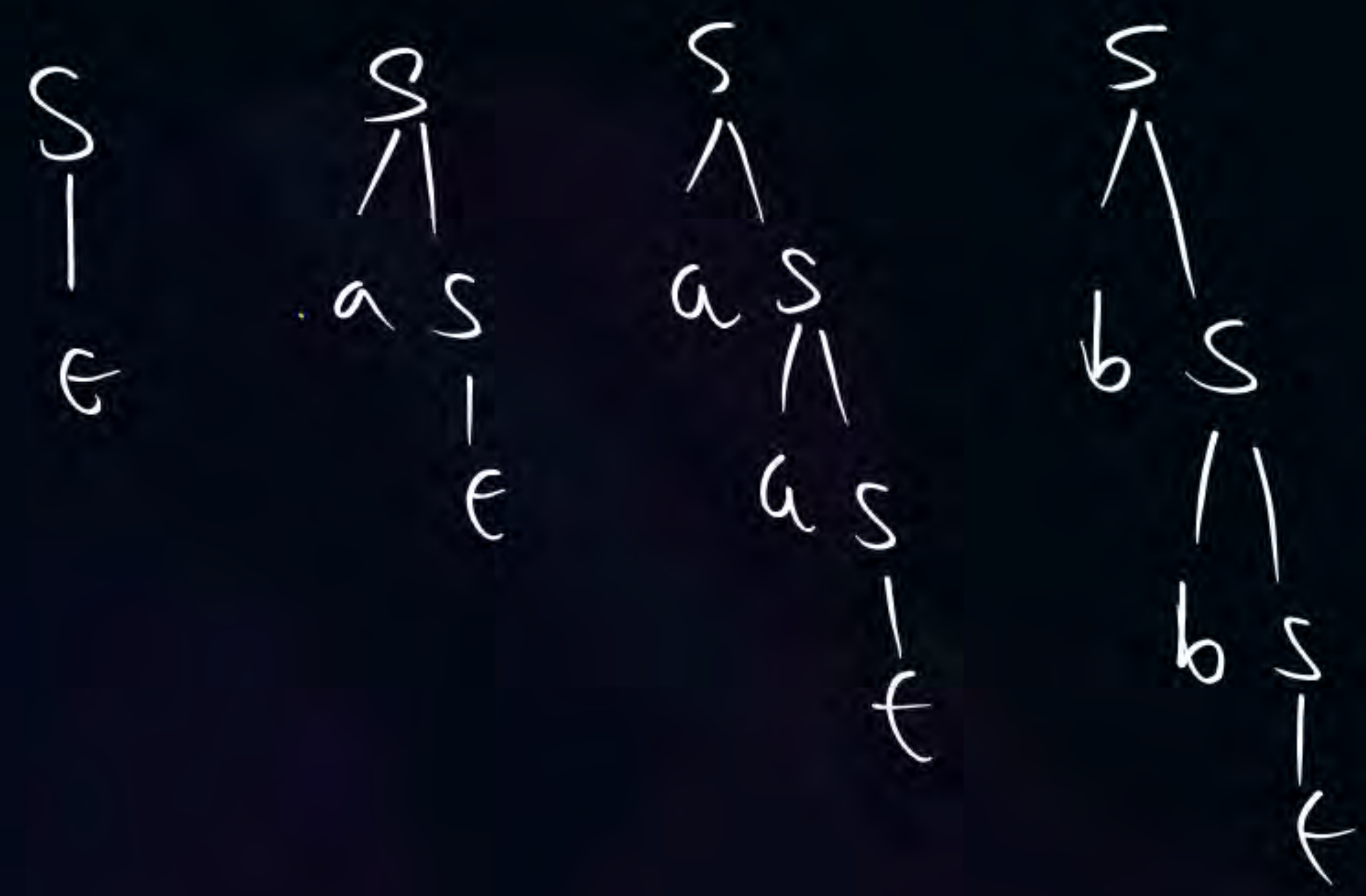
- For Context free language there exist a grammar known as context free grammar.
- For Context sensitive language there exist a grammar known as context sensitive grammar.
- For recursive enumerable language there exist a grammar known as unrestricted grammar.

[NAT]

#Q. Identify language generated by following grammar.

① $\boxed{S \to aS \mid bS \mid \epsilon}$

$L = \{\epsilon, a, a^2, a^3, \ldots, b, b^2, b^3, \ldots, ab, ba, aab, bba \ldots\}$



$(a+b)^*$

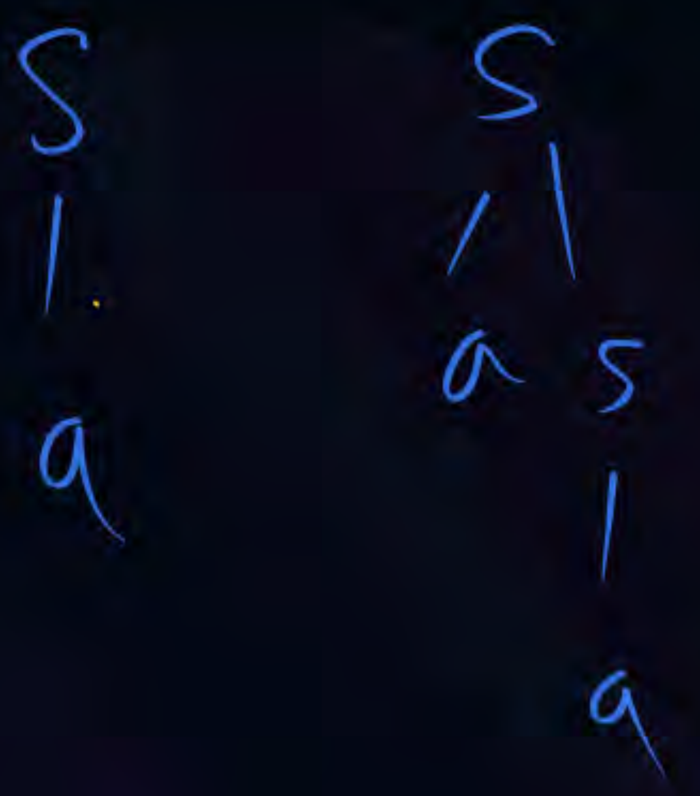
[NAT]

#Q. Identify language generated by following grammar.

2 $\boxed{S \rightarrow aS \mid a}$

$L = \{a, a^2, a^3, a^4, \ldots\}$

a^+

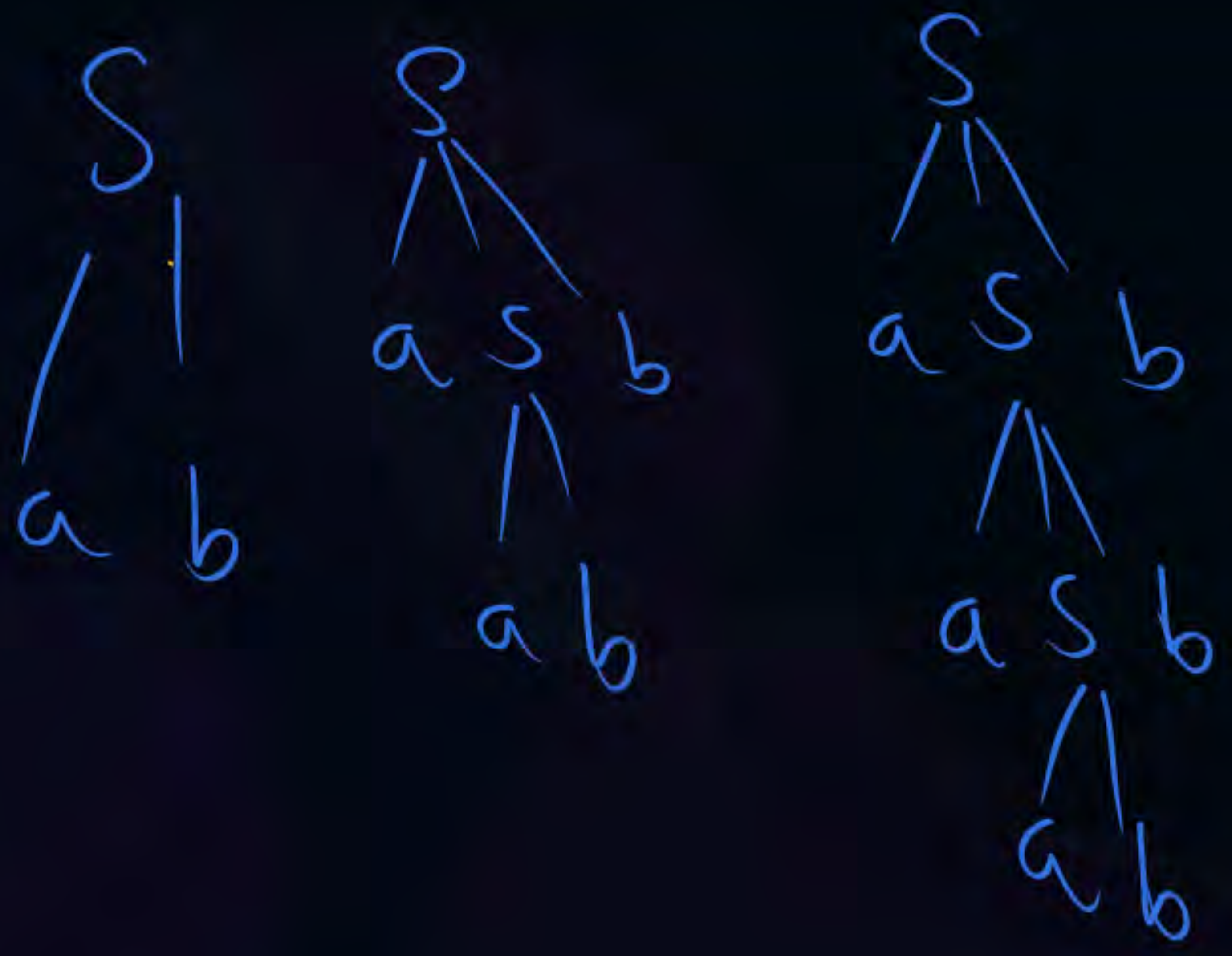


[NAT]

#Q. Identify language generated by following grammar.

$$\boxed{S \to aSb \mid ab}$$

$$L = \{\underline{ab}, a^2\underline{b^2}, a^3\underline{b^3}, \ldots\}$$



$$\{a^n b^n \mid n \geq 1\}$$

[NAT]

$\boxed{\{a^n b^n c^m d^m\}}$

#Q. Identify language generated by following grammar.

$S \rightarrow AB$

$A \rightarrow aAb \mid ab \} \boxed{a^n b^n}$

$B \rightarrow cBd \mid cd \} \boxed{c^m d^m}$

[NAT]

$L = \{WW^R / W \in (a+b)^*\}$

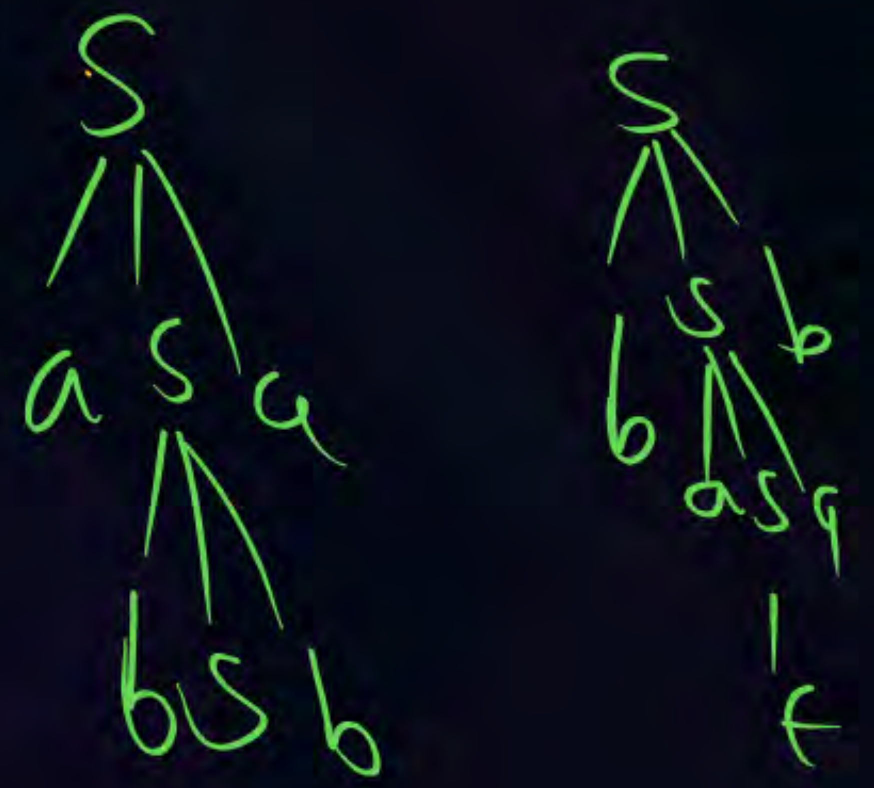
#Q. Identify language generated by following grammar.

$$\boxed{S \to aSa \mid bSb \mid \epsilon}$$

{Even length Palindrome}

$L = \{\epsilon, aa, bb, abba, baba---\}$

$\Sigma = \{a, b\}$



[NAT]

#Q. Construct grammar for the following languages.

$G = (N, T, P, S)$

$\boxed{L = a^* b^*}$

a^* $\quad$ b^*

$$\boxed{\begin{array}{l} S \Rightarrow AB \\ A \Rightarrow aA \mid \epsilon \\ B \Rightarrow bB \mid \epsilon \end{array}}$$

[NAT]

#Q. Construct grammar for the following languages.

$$L = \{ a^n b^n c^m \mid n \geq 1, m \geq 0 \}$$

Terminals

$a^n b^n$

c^*

$$\boxed{\begin{aligned} &{}^{1}\; S \rightarrow AB \\ &A \rightarrow \overset{2}{aAb} \mid \overset{3}{ab} \\ &B \rightarrow \overset{4}{cB} \mid \overset{5}{\epsilon} \end{aligned}}$$

Type of Grammar

$\alpha \to \beta$

Type	Language (Grammers)	From of Productions	Accepting Automata
3	Regular Grammar	R.L.G. $\boxed{A \to xB \mid x}$ Left L.G. $\boxed{A \to Bx \mid x}$	Finite Automaton
2	Context -free Grammar	$A \to \alpha$ $\alpha \in (V+T)^*$	Pushdown Automaton
1	Context sensitive Grammar	$\cancel{A}\,\alpha \to \beta$ $\alpha, \beta \in (V+T)^+$ With $\lvert\alpha\rvert \le \lvert\beta\rvert$	LBA
0	Unrestricted Grammar	$\alpha \to \beta$ $\alpha, \beta \in (V+T)^*$	Turing machine

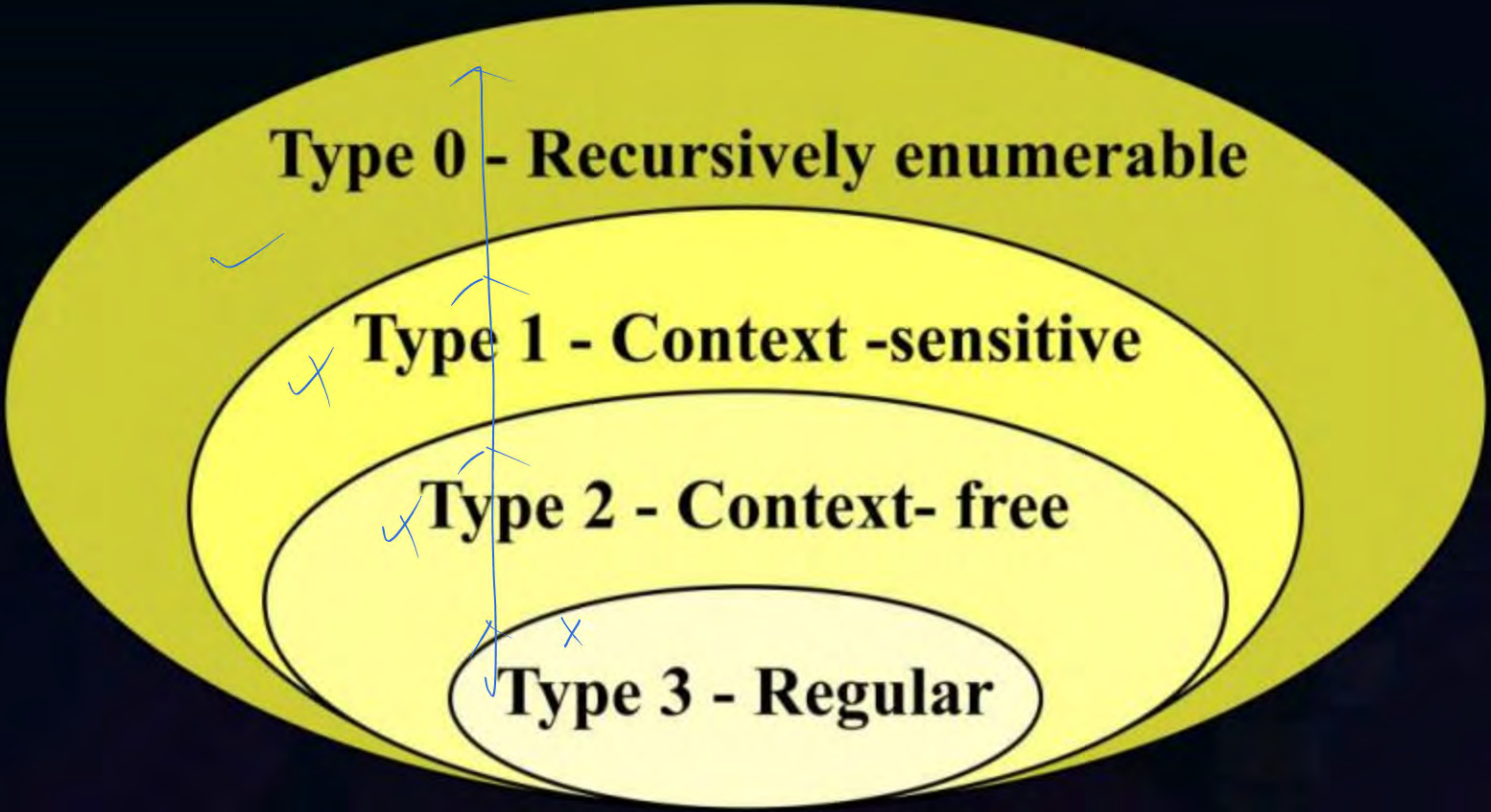
① $\boxed{\begin{array}{l} S \to aAbB \\ aA \to a \\ bB \to b \end{array}}$

② $\boxed{\begin{array}{l} S \to ET \\ E \to a \\ T \to a \end{array}}$

CFG

③ $\boxed{\begin{array}{l} S \to Sa \\ S \to b \end{array}}$ Type 3

Regular Grammar


Type 0 - Recursively enumerable
Type 1 - Context -sensitive
Type 2 - Context- free
Type 3 - Regular

THANK - YOU